ic_autolab round 4 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T03:30:01Z at git `bff904254912195ba9f42027279b0db7816c1b3c`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-6c0998` (`python3 tools/pcc/pcc.py standing PCC-20261104-6c0998`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
28 **improvement**	ksum4-n21	0.654 [0.636, 0.672]	0.611 [0.593, 0.630]	0.416 [0.399, 0.435]	sel < inc	10.939 / 11.226 / 13.809	16.737 / 18.380 / 33.168	1.015 / 1.110 / 1.966	10.8x / 10.1x / 7.0x
32 **no_challenger**	none (best candidate `mitm2-n358`, train S 24.150 / 27.736 / 60.009)	-	-	-	-	-	24.138 / 26.515 / 47.907	0.844 / 0.927 / 1.672	inc: 28.6x / 28.6x / 28.6x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.6536 [0.636, 0.672] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 32: no_challenger

Train (development screen, in sample)

k = 28: session `ECBS1he0b4cc83f712` (run exit 0, verify exit 0, receipt sha256 `e864dd0a6062199aaee28da8dad2c980e9f829092a47be28e4d3f7d240e7b114`)

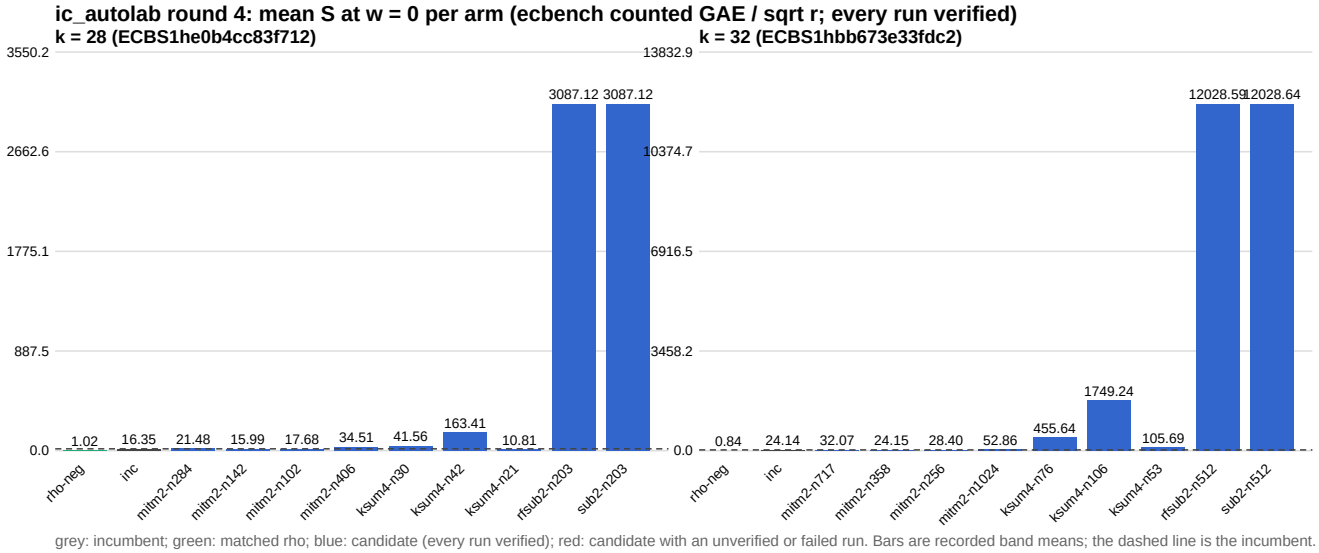
arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	16	16	1.016 / 1.112 / 1.968	matched reference
`inc`	mitm2	203	`prime-projected-abscissa:size=203,scan_cap=3248`	`mitm:negation_folded=1`	16	16	16.350 / 17.914 / 31.997	incumbent
`mitm2-n284`	mitm2	284	`prime-projected-abscissa:size=284,scan_cap=4544`	`mitm:negation_folded=1`	16	16	21.485 / 22.644 / 33.081	incumbent neighbourhood: N = 1.4 x 203
`mitm2-n142`	mitm2	142	`prime-projected-abscissa:size=142,scan_cap=2272`	`mitm:negation_folded=1`	16	16	15.987 / 18.266 / 38.780	incumbent neighbourhood: N = 0.7 x 203
`mitm2-n102`	mitm2	102	`prime-projected-abscissa:size=102,scan_cap=1632`	`mitm:negation_folded=1`	16	16	17.678 / 20.669 / 47.587	incumbent neighbourhood: N = 0.5 x 203
`mitm2-n406`	mitm2	406	`prime-projected-abscissa:size=406,scan_cap=6496`	`mitm:negation_folded=1`	16	16	34.505 / 35.252 / 41.971	incumbent neighbourhood: N = 2 x 203
`ksum4-n30`	ksum4	30	`prime-projected-abscissa:size=30,scan_cap=480`	`ksum:m=4`	16	16	41.559 / 41.670 / 42.668	cross-size transfer: ksum4-n12 is the k = 24 incumbent at N / N_a = 0.148
`ksum4-n42`	ksum4	42	`prime-projected-abscissa:size=42,scan_cap=672`	`ksum:m=4`	16	16	163.411 / 163.522 / 164.524	cross-size transfer neighbourhood:

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
								N = 1.4 x 30 (ksum4 from k = 24)
ksum4-n21	ksum4	21	prime-projected-abscissa:size=21,scan_cap=336	ksum:m=4	16	16	10.814 / 11.076 / 13.431	cross-size transfer neighbourhood: N = 0.7 x 30 (ksum4 from k = 24)
rfsub2-n203	rfsub2	203	prime-projected-abscissa:size=203,scan_cap=3248	rootfind-subtract:m=2	16	16	3087.121 / 3396.478 / 6180.688	exploration: family rfsub2 untried at k = 28, N = 1 x N_a
sub2-n203	sub2	203	prime-projected-abscissa:size=203,scan_cap=3248	subtract	16	16	3087.119 / 3395.718 / 6173.109	exploration: family sub2 untried at k = 28, N = 1 x N_a

k = 32: session `ECBS1hbb673e33fdc2` (run exit 0, verify exit 0, receipt sha256 `0792eae67547acfb35a46ae1b9a432c2bb3801530da545f67f441f9479a14996`)

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	-	-	rho.negation	16	16	0.844 / 0.927 / 1.672	matched reference
inc	mitm2	512	prime-projected-abscissa:size=512,scan_cap=8192	mitm:negation_folded=1	16	16	24.138 / 26.515 / 47.907	incumbent
mitm2-n717	mitm2	717	prime-projected-abscissa:size=717,scan_cap=11472	mitm:negation_folded=1	16	16	32.069 / 33.793 / 49.315	incumbent neighbourhood: N = 1.4 x 512
mitm2-n358	mitm2	358	prime-projected-abscissa:size=358,scan_cap=5728	mitm:negation_folded=1	16	16	24.150 / 27.736 / 60.009	incumbent neighbourhood: N = 0.7 x 512
mitm2-n256	mitm2	256	prime-projected-abscissa:size=256,scan_cap=4096	mitm:negation_folded=1	16	16	28.404 / 33.414 / 78.499	incumbent neighbourhood: N = 0.5 x 512
mitm2-n1024	mitm2	1024	prime-projected-abscissa:size=1024,scan_cap=16384	mitm:negation_folded=1	16	16	52.863 / 54.035 / 64.586	incumbent neighbourhood: N = 2 x 512
ksum4-n76	ksum4	76	prime-projected-abscissa:size=76,scan_cap=1216	ksum:m=4	16	16	455.642 / 455.751 / 456.731	cross-size transfer: ksum4-n12 is the k = 24 incumbent at N / N_a = 0.148
ksum4-n106	ksum4	106	prime-projected-abscissa:size=106,scan_cap=1696	ksum:m=4	16	16	1749.242 / 1749.445 / 1751.274	cross-size transfer neighbourhood: N = 1.4 x 76 (ksum4 from k = 24)
ksum4-n53	ksum4	53	prime-projected-abscissa:size=53,scan_cap=848	ksum:m=4	16	16	105.689 / 105.776 / 106.560	cross-size transfer neighbourhood: N = 0.7 x 76 (ksum4 from k = 24)
rfsub2-n512	rfsub2	512	prime-projected-abscissa:size=512,scan_cap=8192	rootfind-subtract:m=2	16	16	12028.586 / 13232.542 / 24068.140	exploration: family rfsub2 untried at k =

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
sub2-n512	sub2	512	prime-projected-abscissa:size=512,scan_cap=8192	subtract	16	16	12028.643 / 13231.431 / 24056.523	exploration: family sub2 untried at k = 32, N = 1 x N_a



train screen

Selection

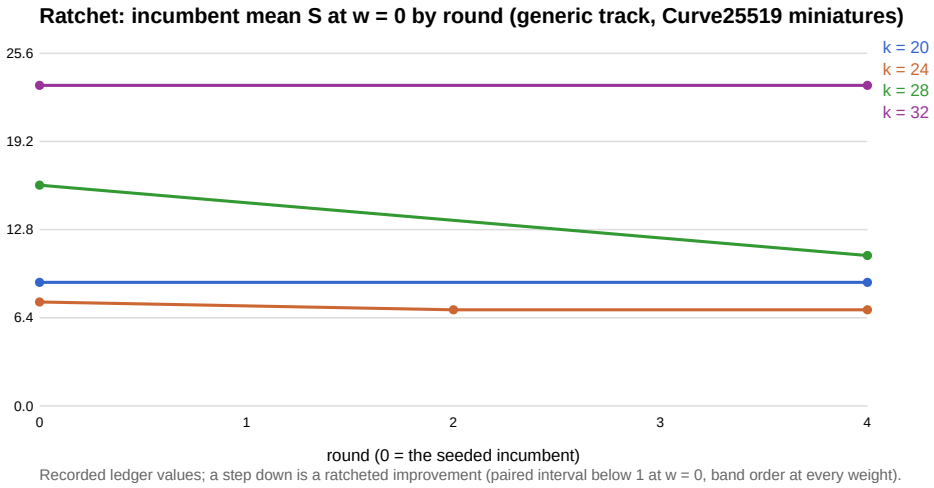
- k = 28: challenger ksum4-n21 (Pareto alternative at w = 1: ksum4-n21)
- k = 32: no_challenger (Pareto alternative at w = 1: mitm2-n717)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
28	ECBS1h4fa56ebd629d	0	0	6b1c0a3f7fc28a45c31c7bb9a197921ee1a4efff121e3eec87243d5396c945bd	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	ksum3-n11 (prime-projected-abscissa:size=11,scan_cap=176 + ksum:m=3)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	ksum4-n12 (prime-projected-abscissa:size=12,scan_cap=192 + ksum:m=4)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	ksum4-n21 (prime-projected-abscissa:size=21,scan_cap=336 + ksum:m=4)	4	10.939 / 11.226 / 13.809	10.8x / 10.1x / 7.0x
32	mitm2-n512 (prime-projected-abscissa:size=512,scan_cap=8192 + mitm:negation_folded=1)	0	23.310 / 25.520 / 45.410	24.7x / 24.6x / 24.3x



ratchet

Evidence and deviations

- ecbench binary sha256: `3fd22853485e388782ac0821295c56f81725fc832fd3610978be8088b5098bca`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.